Phosphosite coverage

How many sites each sample quantified, split by how confidently they are localised.

AT A GLANCE

Where	Phospho → Phosphosite Coverage Plot
Data level	Phosphosite. The page reads the phospho matrix as it was uploaded, so filtering and imputation performed elsewhere do not change these bars.
Needs	A phospho dataset and its annotation. Conditions and sample order come from the annotation.
Classes	Class I is PTM_localization ≥ 0.75, Not Class I is below it, and a third class, Unclassified, holds the rows whose localisation is missing or outside $[0, 1]$. The 0.75 is fixed in the code and follows no control.
Two modes	**Type** Normal draws one stacked bar per sample; Summary draws one bar per class per condition, with the standard deviation and the individual samples over it.
Table	Yes, below the plot, downloadable as CSV. Its columns change with **Type**.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

The ordinary coverage plot asks whether every sample worked. This one asks the same question and one more: of the sites a sample quantified, how many are pinned to a particular residue. Two datasets can produce identical bar heights and differ entirely in how much of that height can be attributed to a site rather than to a peptide.

Read both the total height and the localisation categories. Comparable totals describe similar detection depth, not proof of equally good runs. The Class I fraction concerns confidence in the assigned residue; it can also depend on the peptides detected, their spectra and the search setup. A different fraction is a reason to inspect those factors, not a diagnosis of fragmentation failure or a requirement to make all fractions equal.

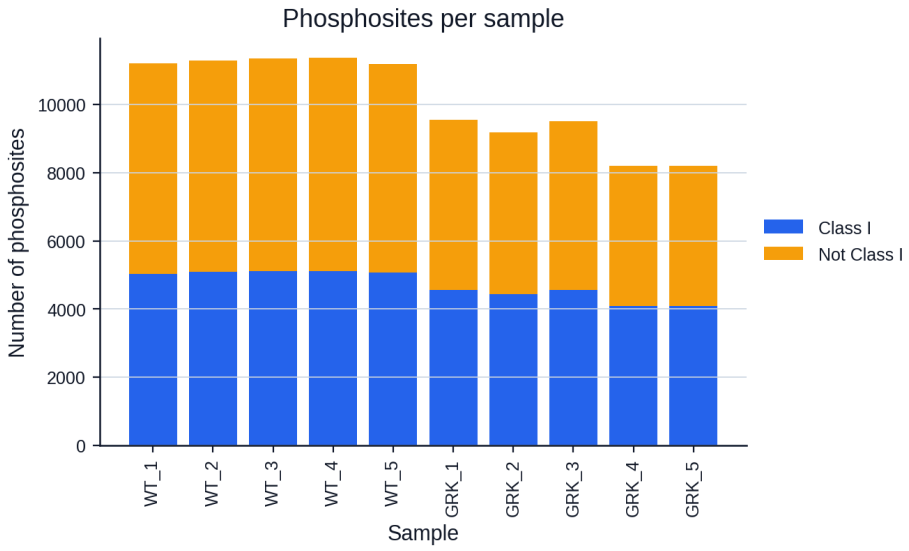


Figure 1.1. Per-sample mode on the example dataset: five WT and five GRK samples, Class I sites forming the lower blue segment of each bar and Not Class I the orange one above it, with the total height the number of phosphosites quantified in that sample. The five GRK bars are noticeably shorter than the five WT bars, in the Class I segment as well as in the total. No Unclassified segment is drawn, because every row of this dataset carries a usable localisation value.

Reading it

- The lower segment: sites in that sample whose PTM_localization is at least 0.75 and that carry a value in that column.
- The segment above it: sites with a localisation below 0.75, counted the same way.
- The full height of the bar: every site quantified in that sample, whatever its localisation.
- A third segment, slate grey, appears above the other two only if the dataset contains rows with no usable localisation value. It is labelled Unclassified in the legend, and its absence means every row carried a localisation between 0 and 1.
- The label under each bar: condition_n, numbered from 1 within each condition rather than across the experiment. With **Toggle ID** on, the sample's number from the original file name is added in brackets on a second line.

EASY TO GET WRONG

Class I is a statement about *where* the phosphate sits, not about how well the site was measured. A Class I site can have a poor quantitative signal, and a Not Class I site can be measured precisely; what is uncertain is which of two or three candidate residues on the peptide carries the modification. Treat the split as a guide to how far a site-level conclusion can be pushed — a regulated Not Class I site is a regulated *peptide* — and not as a quality filter on the intensities.

2 Controls

Controls on this page

Control	What it does	Default	Where
Type	Normal: one stacked bar per sample. Summary: one group of bars per condition, each class its own bar, with mean, standard deviation and the individual samples.	Normal	page
Conditions	Which conditions to draw. Samples of the conditions not selected are left out of the figure and the table altogether. Selecting none is the same as selecting all.	all conditions	page
Toggle ID	Adds the sample's number, taken from the original file name, under each label.	off	page
Random Seed	Summary only. Fixes the horizontal jitter of the individual sample points so the same figure is reproduced exactly.	69	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page.	20 × 10, 300	page
Download Plot	Exports the figure in the format chosen in the dropdown beside it: PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Class I	The colour of the Class I bar or segment. Until one is chosen the plot draws its own blue, #2563eb.	#2563eb	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Not Class I	The colour of the Not Class I bar or segment. The plot's own default is amber, #f59e0b; the picker opens on #94a3b8. The Unclassified segment is always slate grey and has no picker.	#f59e0b	appearance
Value labels	Size of numbers printed on bars. This plot prints none, so the field has nothing to act on — see the open question below.	8	appearance
Title	Shown or hidden, and its text, size, font and colour. The plot writes Phosphosites per sample, or Phosphosites per condition in summary mode.	shown	appearance
Legend	The class legend, drawn to the right of the plot. Turning it off also returns the width it was occupying.	shown	appearance
Condition labels	Renames a condition on the axis in summary mode without renaming it anywhere else.	the annotation's names	appearance
Axis frame, grid, background	Which spines are drawn, the grid over them, paper colour and the preset themes.	default	appearance

The panel shows no **Value labels** card here, because these bars carry no printed counts for it to size.

> **WORTH KNOWING**
>
> The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

> **HOW IT IS COMPUTED**
>
> PTM_localization is read as a number and any value outside [0, 1] is discarded, which puts that row with the ones that had no value at all. Each row then belongs to exactly one of three classes: Class I at ≥ 0.75, Not Class I below it, Unclassified for the rest.

Every sample column is read as a number, and a value counts as present when it is finite. If the annotation is not log2-transformed, a zero is additionally treated as missing, because a report that writes 0 for an unmeasured phosphosite would otherwise inflate every bar. For sample j and class c over the m rows of the matrix,

$$\text{count}_{jc} \;=\; \sum_{i=1}^{m} \mathbf{1}[\, i \in c \,]\; \mathbf{1}[\, x_{ij} \text{ present} \,] \,.$$

The three counts are stacked in the order Class I, Not Class I, Unclassified, so the height of the bar is their sum: the number of phosphosites quantified in that sample.

SUMMARY MODE

The per-sample counts are computed exactly as above and then aggregated within each condition. Each class gets its own bar, the height being the mean of the counts of the samples of that condition and the error bar their standard deviation over $n - 1$ degrees of freedom. Every sample is additionally drawn as a black point over its bar, displaced sideways by a small random amount that **Random Seed** fixes. A condition with one replicate has no spread; its error bar is reported as zero rather than omitted.

Edge cases

- A sample that quantified nothing draws a bar of height zero rather than disappearing from the axis, so the gap is visible.
- The Unclassified segment is drawn only when at least one such row exists. Its absence from the legend is therefore informative.
- Sites that carry a localisation but no value in a given sample are counted in no bar of that sample; they are still counted in the samples that did quantify them.
- Deselecting a condition in **Conditions** removes its samples from the figure and from the table, and renumbers nothing: the remaining samples keep the condition_n labels they had.
- This page has no data-source selector. It always reads the matrix as uploaded, so a filter or an imputation applied on the Data page leaves these bars where they were.

4 The table below the plot

In Normal mode:

Columns in this table

Column	What it holds
Sample	The sample, labelled condition_n as on the plot.
Class I	Sites with localisation ≥ 0.75 quantified in that sample.
Not Class I	Sites with localisation below 0.75 quantified in that sample.
Unclassified	Sites with no usable localisation quantified in that sample.
Condition	The condition the sample belongs to.
Total	The three counts added — the height of the bar.

In Summary mode the table is replaced by one row per condition, holding Condition and then a mean and a standard deviation for each of the four quantities above: classIMean and classISd, notClassIMean and notClassISd, unclassifiedMean and unclassifiedSd, totalMean and totalSd. A standard deviation that cannot be computed is written as 0.0.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = phossite_coverage_plot(data3, meta2, id=False)
```

R

```
fig <- phossite_coverage_plot(data3, meta2, id = FALSE)
```

The stacking, the Class I boundary at 0.75 and the Unclassified segment are the same as on the page, and phossite_coverage_plot_summary draws the condition-level version. One difference is in the labels: the notebook function numbers the samples of a condition from zero, so its first bar is WT_0 where the page writes WT_1.

6 Exporting

The download reproduces the plot at the width, height and resolution set on the page; the figure on screen is fitted to the window instead. PDF for a figure that will be scaled or edited, PNG for a slide. The table underneath downloads separately as CSV, in whichever of its two shapes **Type** is currently producing.

Investigate a noticeably different run against its experimental peers. Lower coverage can have biological, acquisition or processing explanations; a lower Class I fraction can reflect a different peptide mixture as well as localisation evidence. Unclassified means a usable probability was not supplied for those observations. Check the source column and export format, but do not invent a localisation class for missing probabilities.

Behind this page

Function	What it produces
phossite_coverage_plot	The per-sample stacked plot, in the Python package
phossite_coverage_plot_summary	The condition-level summary, with spread and individual samples
phospho_coverage_table	The table under the plot, in either shape
phospho_coverage_png	The figure the page draws